

An Application of Tension Splines in Swap Curve Interpolation

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Abstract

In what follows we outline how to configure and use the tension feature applied to the FRA instruments on the Standard Swap Curve (STD Curve) in M-Library.

Introduction

Tension splines offer a blended combination of linear interpolation and cubic spline interpolation by means of a penalty function. In the extremes a high penalty converges towards linear interpolation and a low penalty converges to cubic spline interpolation. Cubic spline functions can over oscillate and overfit the target curve; tension splines are typically used to overcome this undesired feature.

Tension Implementation

Tension has been applied to the FRA calibration instruments of the swap curve by inserting artificial points, which have been linearly interpolated between sequential forward rate calibration data. Inserting daily points i.e. tension gap = 1 has the effect of transforming the functional form of the yield curve to linear. A less frequent creation of synthetic points (higher tension gap) relaxes this transformation and tends to a cubic spline functional form.

STD Swap Yield Curve Parameters

In M-Library the function *mirSetUpSwapCurve* builds the standard swap curve. The following parameters are used to control Swap curve interpolation: *yieldgen.interpolation*, *ApplyTension* and *TensionGap*. These STD swap yield curve parameters can be found in the curve properties and FRA conventions tables as shown in (1).

The *yieldgen.interpolation* flag recommended setting is *fn_splineinterpolation*, however a useful alternative is *fn_linearinterpolation*. The *ApplyTension* parameter controls if tension is applied to FRA instruments. Setting this to true will dampen and reduce spline oscillation and curve overfitting. It is recommended to set this parameter to true. Finally the *TensionGap* parameter controls the amount of tension to apply. The recommended setting is 2, representing a tension gap of 2 business days.

Forward Rate Parameters

For forward rate calculation in M-Library can be generated using the *mirGetForwardRate2* function kernel and forward rate interpolation can be controlled using the *interpolation* and *interpForwards* parameters. The *interpolation* parameter can be set to linear or spline. The *interpForwards* parameter if set to true will interpolate forward rates between pillar points and if set to false will interpolate on discount factors. Interpolation on for-

Curve Properties

AsOfDate	04-Dec-15
Currency	EUR
basis.interpolation	fn_splineinterpolation
basisfunction	fn_basisfunc2
daycount	Act/365_isda
frequency	continuous
future.interpolation	fn_splineinterpolation
interpolation	fn_splineinterpolation
isadjustdf	TRUE
isfutureuse	FALSE
isspotuse	FALSE
isswaptenoradjust	FALSE
slidingrule	no_change
yieldgen.interpolation	fn_splineinterpolation
isfrause	TRUE
basis.interpolation.3m6mbasis	fn_splineinterpolation
basis.interpolation.xccybasis	fn_splineinterpolation
isRenotionalAdjust	TRUE
dfcurvename	OIS

FRA Conventions

Calendar	TGT
DayCount	ACT/360
ResetLag	2
SlidingRule	MOD_FOLLOWING
IsEomRoll	FALSE
ApplyTension	TRUE
TensionGap	2

Figure 1: STD Swap Curve, Curve Parameters

wards is the preferred and recommended setting. Forward rate parameters are shown in figure (2) below.

Results

Tension results are demonstrated visually in figure (3). Two large artificial shifts were added to FRA calibration instruments. In red and labelled snapshot we can see how a cubic spline overfits the curve and over-oscillates. Applying tension with a 2 day tension gap is shown in blue and labelled EUR6ML.

Conclusion

The tension spline feature reduced curve overfitting and provides a better reflection of market curves. Accordingly it is recommended that the ApplyTension parameter is set to true with a TensionGap of 2 business days.

References

- [1] **Andersen, L** and **Piterbarg, V** (2010) Textbook: Interest Rate Modeling Volume I-III, Atlantic Financial Press
- [2] **Andersen, L** (2005) "Yield Curve Construction with Tension Splines" SSRN Social Science Research Network (<http://ssrn.com/abstract=871088>)

Select Curve & Interpolation

CurveID	EURYC
Interpolation	Spline
InterpForwards	TRUE

Forward Curve Chart Settings

Forward Curve	6M
Display Years	2

ApplyTension	TRUE
TensionGap	2

Figure 2: Forward Rate Parameters

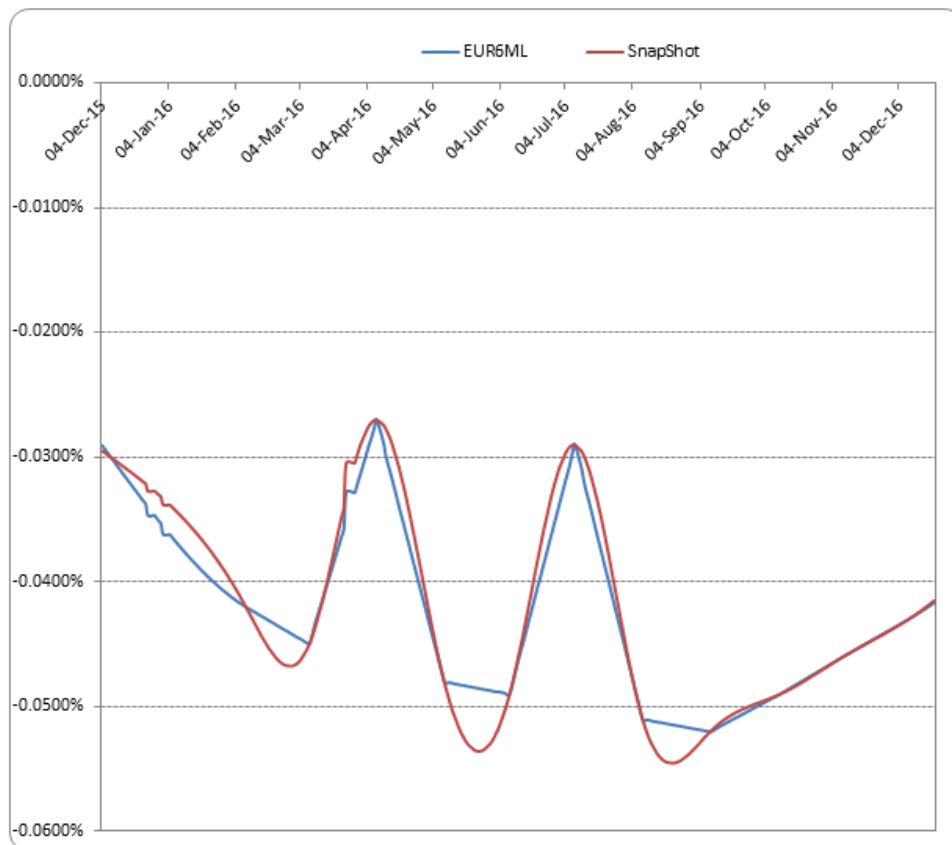


Figure 3: Forward Rate Chart